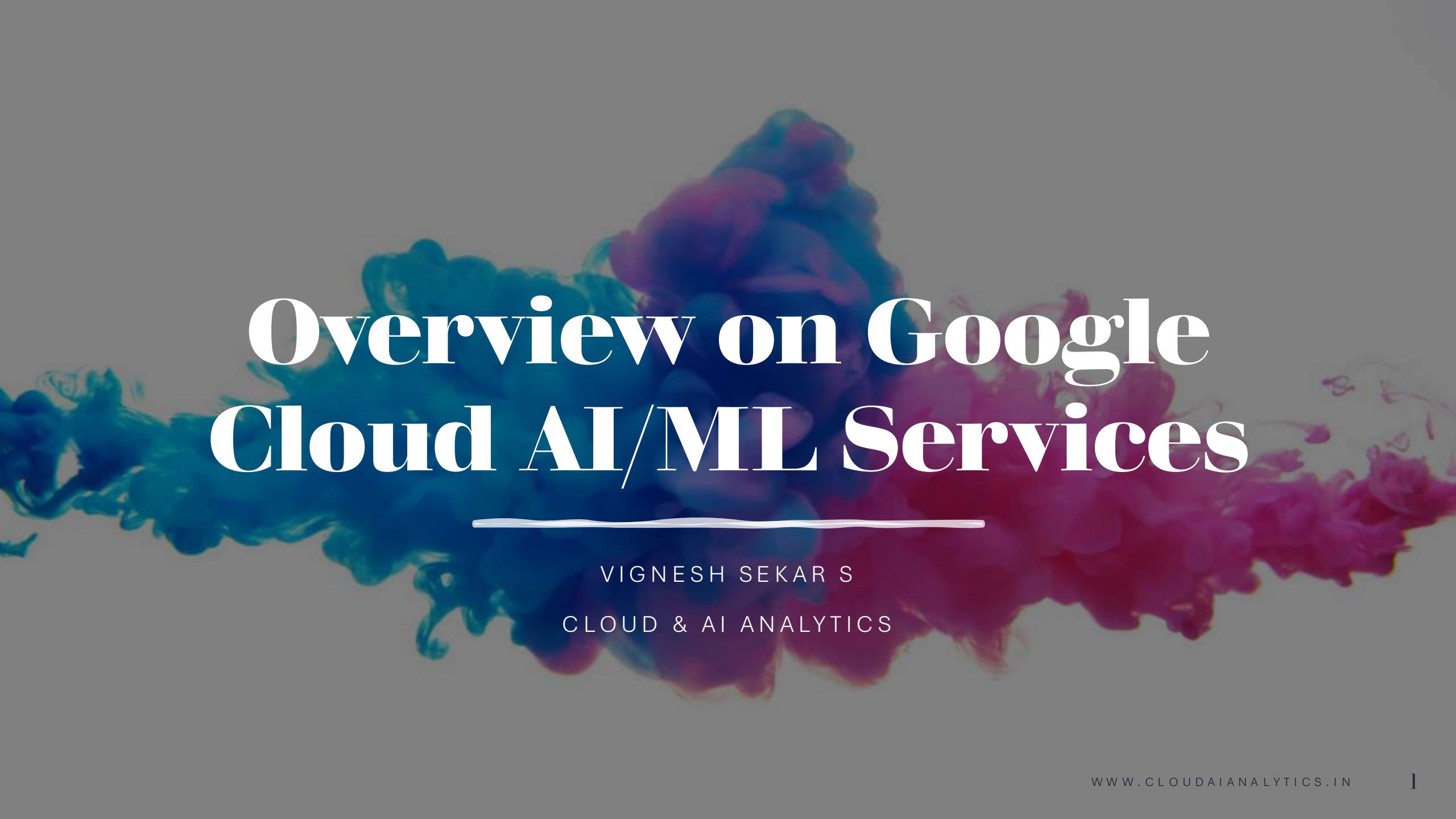




Overview on Google Cloud AI/MIL Services

VIGNESH SEKAR S
CLOUD & AI ANALYTICS

Overview on Machine Learning

- Machine learning is a subfield of **artificial intelligence (AI)**.
- Rapidly evolving with potential to revolutionize many aspects of our lives.
- It involves creating Machine Learning Algorithms
- To make decisions or predictions without being explicitly programmed.

Overview on Machine Learning

- Function of Machine Learning
 - Descriptive
 - Predictive
 - Prescriptive

Happy Learning ..!

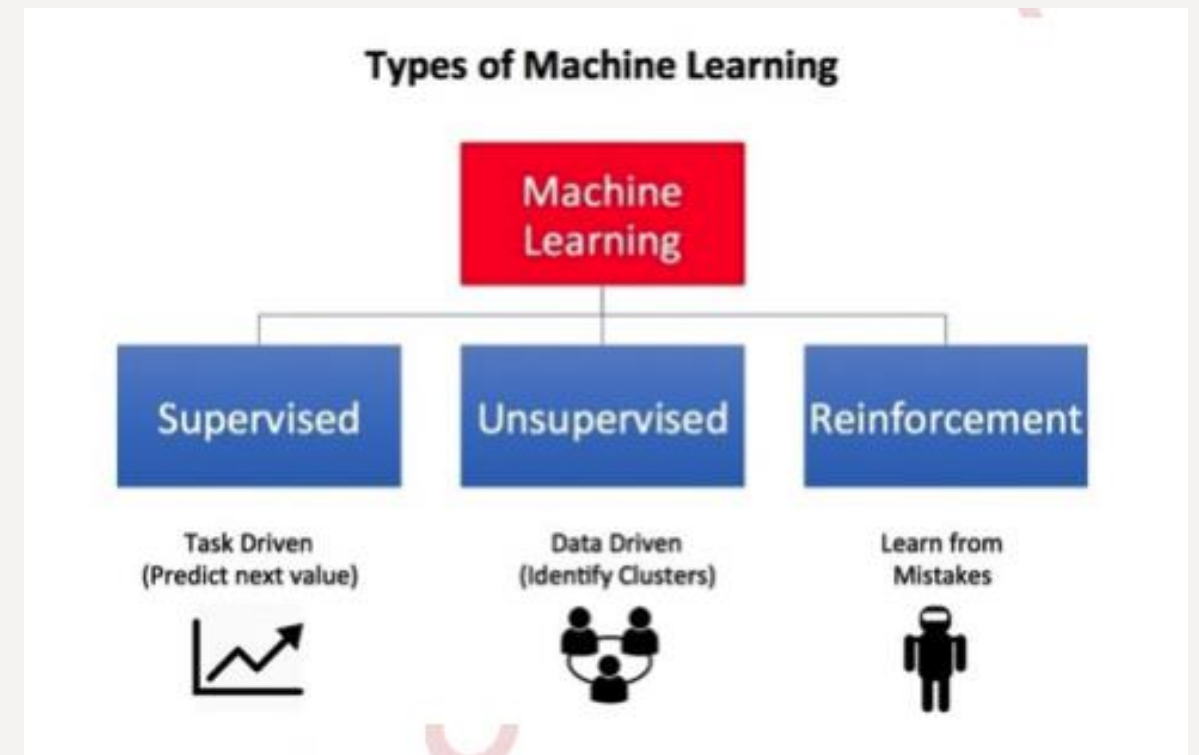
Components of Machine Learning

- The Key components of Machine learning are
 - Learning from Data
 - Algorithms
 - Models
 - Training and Testing
 - Predictions
 - Evaluation

Happy Learning ..!

Types of Machine Learning

- The types of Machine learning are
 - Supervised Learning
 - Unsupervised Learning
 - Semisupervised Learning
 - Reinforcement Learning



Happy Learning ..!

Overview on Machine Learning products on Google Cloud Platform

- Google Cloud offers a suite of machine learning products designed to help businesses of all sizes leverage the power of AI.
- **Vertex AI**
 - Vertex AI is a unified platform for building, deploying, and managing machine learning models.
- **Auto ML**
 - Automates the process of building machine learning models

- AutoML Vision
- AutoML Natural Language
- AutoML Tables
- **Pre-trained AI APIs**
 - Natural Language API
 - Vision API
 - Speech to Text
 - Text to Speech
 - Translation API

Happy Learning ..!

Overview on EDA

- It involves a detailed investigation of the dataset to understand
 - Characteristics
 - Identify patterns and Relationships
 - Identify Data Quality issues
 - Visualize data
 - Uncover potential issues.
- EDA helps you gain insights into the data before proceeding with further analysis or modeling.

Overview on EDA

- The common techniques used are
 - Statistics
 - Data Visualization
 - Data Transformation
- Tools used for EDA are
 - Programming languages - Python, R and SQL
 - Interactive environments – Jupyter Notebooks and RStudio

Happy Learning ..!

Overview on Feature Engineering

- It is the process of selecting, extracting, and transforming relevant features from raw data to enhance the performance of machine learning models.
- Common feature engineering techniques are
 - Scaling and Normalization
 - Handling Categorical values
 - Dimensionality Reduction

Happy Learning ..!

Benefits Of GCP's Machine Learning products

- The benefits of GCP's Machine learning products are
 - Scalability and Performance
 - Ease of use
 - Integration and Innovation
 - Cost Effectiveness
 - Security

Happy Learning ..!

Applications Of Machine Learning products

- Machine learning are used in wide variety of applications such as
 - Healthcare
 - Finance
 - Retail
 - Transportation
 - Recommendation system

Happy Learning ..!

Challenges in Machine Learning

- Machine learning faces several challenges such as
 - Data challenges
 - Algorithmic challenges
 - Computational challenges
 - Ethical challenges

Happy Learning ..!

Overview on BigQuery ML

- It allows users to build and execute machine learning models directly within BigQuery, Google's data warehousing service.
- Using SQL queries
- Features of BigQuery ML are
 - ML for structured data analysis
 - Supports various ML models
 - Integrates with GCP ecosystem
 - Data Analysts can build ML models

Lab on Cloud BigQuery ML

Happy Learning ..!